Software Requirements Specification

for

ProQuiz

Version 0.1 Alpha

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Revision History

Name	Date	Reason For Changes	Version
ProQuiz A1	21/03/2023	First development version	0.1 Alpha

1. Introduction

1.1 Purpose

The purpose of this document is to present a detailed description of the software ProQuiz. It will explain the purpose and features of the software, the interfaces of the software, what the software will do and the constraints under which it must operate. This document is intended for users of the software and also potential developers.

1.2 Document Conventions

This Document was created based on the IEEE template for System Requirement Specification Documents.

1.3 Intended Audience and Reading Suggestions

- Users who want to use the application to participate in quizzes which are conducted by other users or other openly available quizzes on the application.
- Users who want to create their own quizzes for other users to participate in.
- Programmers who are interested in working on the project by further developing it or fixing bugs.

1.4 Product Scope

ProQuiz is a web application that people can use to create their own personal quizzes, using their own questions or optionally from the in-built question bank. Users can participate in such a quiz event by getting accepted by the quiz maker, or participate in general open quizzes provided by the application. Users will also have the option to contribute to the question bank by providing relevant questions and answers.

1.5 References

ProQuiz's GitHub page: https://github.com/srdevanarayan/ProQuiz
IEEE Template for SRS Documents: https://goo.gl/nsUFwy

2. Overall Description

2.1 Product Perspective

ProQuiz is developed for everyone who wants a simple quiz application in which anyone can make and participate in quizzes. It will have various customization options, and secondary features such as an inbuilt question bank. This application is not unique in terms of the idea, but is an effort to create an effective system for everyone to gain and test their knowledge with minimal effort. It is developed as a web application, and hence can run on almost all operating systems & devices such as Windows, Mac OS, Linux, Android, iOS etc. having modern versions of web browsers.

2.2 Product Functions

- Login: Login to the application as a registered user.
- Register: Create a new account to use the application.
- Join a custom quiz: Join a quiz event by providing the unique quiz code. The quiz maker has to approve the quiz taker.
- Join a general/public quiz: Join a public quiz event under various categories (eg: Computer science, Biology etc.) and subcategories (eg: DSA, Human body etc.)
- View history of quizzes participated in/ quizzes conducted.
- Create a quiz event: Create a new custom quiz event with the help of existing question bank or your own questions. Options to set parameters such as time limit for each question.
- Approve quiz takers: Approve the quiz takers who can participate in the quiz event. There will be options to filter out the requests for participation.
- View results of quizzes for both quiz maker and quiz taker.
- Start quiz manually or automatically, set end time for the quiz or do it manually.
- Support: Option is provided to seek support from the developers in case of any problems.

Note: These may change as the software is being implemented. New functions may be added or some functions maybe modified to reflect the changing requirements and constraints.

2.3 User Classes and Characteristics

- Quiz takers: Users who want to participate in a custom quiz event or a general quiz, such as students.

- Quiz makers: Users who want to create a quiz event so that other quiz takers can participate in it, such as teachers.
- Programmers who are interested in working on the project by further developing it or fix existing bugs

2.4 Operating Environment

Any OS or devices such as Windows, Linux, MacOS, Android, iOS etc. are supported provided that there is a modern browser such as Google Chrome, Firefox, Edge etc. installed.

2.5 Design and Implementation Constraints

ProQuiz is a web application developed with the help of MERN stack technology, using MongoDB, Express.js, React and Node.js. The core programming language is JavaScript. The application is designed in an intuitive and simple way such that users can use it without having advanced technical knowledge. Design is in a way so that there is focus on only one or few things at a time, keeping the user interface uncluttered. The application is responsive and fast enough to load on low end devices and poor internet connectivity.

2.6 User Documentation

There is ReadMe file in ProQuiz's GitHub repository: ProQuiz/README.md at main · srdevanarayan/ProQuiz (github.com)

2.7 Assumptions and Dependencies

ProQuiz is mainly developed using JavaScript and its libraries and frameworks. Hence it requires that users use browsers that supports modern JavaScript. For users accessing the application from their personal computers, working input devices such as mouse and keyboard is required. For users accessing the application from their mobile devices, touchscreen input must be in a working condition with no anomalies like ghosting. It is assumed that users have a basic knowledge to navigate the application.

3. External Interface Requirements

3.1 User Interfaces

Even though there are mainly 2 types of users (quiz taker & quiz maker), the user interface is common for both since it enables anyone to act as any type of user.

1. Main Screen: It has fields to enter username and password for login. In case of new account registration, the fields will change to reflect the information that should be entered in order to create a new account.
2. Dashboard: After login, user will see options to select whether they want to act as a Quiz maker or Quiz taker. The page will then change to respective view.
3. Quiz taker screen: There will be options to join a new quiz and view history. 'Join a new quiz' will give options to user so that they can select if they want to join a custom quiz by entering the quiz code or participate in a general quiz by choosing the category and subcategory. 'View history' will give details of past quizzes attended and their results.
4. Quiz maker screen: There will be options to make a new quiz event, quiz events created but not completed, and completed quiz events. New quiz event can be created, and user can include their own questions or with the help of existing question bank. In quiz events created but not completed, users will have the facility to approve quiz takers that are interested to participate in the quiz. In quiz events completed, users can view the results of quiz events that were held previously.
5. Common to both user views, there will options to logout and contribute to the question bank. Logout will sign out the users from their current session. Contribute to the question bank can be used to add your own questions and answers to the question bank that may be used by other users, or to vote the correctness of the existing questions in the question bank.

This is a high-level description of the user interface of the application. This may change during the detailed designing stage during production. A more detailed user interface design details will be available in the final design documents.

3.2 Hardware Interfaces

As ProQuiz is a web application, there is no demanding hardware requirements. The device must have enough memory (RAM) and CPU power to run the browser smoothly, working keyboard and mouse in case of personal computers and touchscreen display in case of mobile devices. Preferably the screen must support colors for better aesthetics.

3.3 Software Interfaces

ProQuiz requires a modern browser that supports JavaScript and its libraries and frameworks. Local storage maybe used to save the user session for a period of time. In the backend, user data and all other data of the applications is stored in a MongoDB database. No interfering plugins or extensions should be used in the browser.

3.4 Communications Interfaces

ProQuiz, being a web application, requires constant internet connection to function. This is essential so as to fetch and update data from/to the database. Without an active internet connection, users cannot use the application.

4. System Features

This section demonstrates ProQuiz's most prominent features.

4.1 User Authentication

Users can create their own account providing necessary details, then login to use the application. All the credentials are stored securely.

4.2 Multiple User Views

A user only need to create a single account to use the application as both Quiz Maker and Quiz Taker. There are options to switch between the user views.

4.3 Participate in both Custom & General Quiz

Users can participate in a custom quiz by providing the quiz code, or participate in a general quiz using the questions from existing question bank. This provides flexibility to users.

4.4 Peer Verified Question Bank

ProQuiz has a public question bank that anyone can contribute to. The correctness of the questions in the question bank is verified by other users by rating. Users can decide which questions to use by checking the rating of questions.

4.5 Flexibility in Creating a Custom Quiz Event

Quiz Makers can create a custom quiz event by adding their own questions or using questions from the question bank. There are many customization options including but not limited to: Manual or automatic starting or ending of a quiz event, individual parameters for each question like time limit etc.

4.6 Multiple Themes

Users can select their own theme for the application according to their own preference.

4.7 View History

There is option to view history of all the quizzes a quiz taker has participated in, quiz events a quiz maker has created, the questions a user has contributed to the question bank. Anyone can view the results of past quizzes if they want to.

5. Other (Nonfunctional) Requirements

5.1 Performance Requirements

ProQuiz must be able to run on low-end devices without affecting its functionality. Software must be designed in a way that it is able to run effectively on devices with low ram and CPU power. Any graphics and other resources used in the software must be optimized to load fast. Application must be responsive so as to work effectively across all kinds of devices. Accessibility must be kept in mind, and features such as providing titles for images, ability to navigate using keyboard etc. must be implemented.

5.2 Safety Requirements

To ensure that no one of ProQuiz's users loses any data while using the application, developer team should update the software and monitor the state of database regularly. There is an option for users to report any bugs in the software. There should be a clear and concise privacy policy that outlines how user data is collected, stored and used.

5.3 Security Requirements

There are no restrictions on which kind of users can use the application. Only users that have an account and signed in should be able to access the application. The credentials and user data must be to be stored in a way that is secured in the database using encryption. Question bank must be monitored to ensure that no undesirable questions is present by giving options to users to flag the questions if needed. Only developers must have access to sensitive information regarding the application and database. Regular security testing is required to ensure no vulnerabilities exists, and if found to fix them.

5.4 Software Quality Attributes

ProQuiz provides the users with a simple and intuitive interface. The main focus of the application is to provide a way for users to test their own or other users' knowledge. The application must remain simple with the interface kept intuitive and uncluttered. No advertisements or any kind of promotional content should be allowed, neither any distracting elements or elements that may slow down the loading and functioning of the application. The application should be reliable and available for users when they need it. This can be achieved through proper system design and testing, as well as regular monitoring and maintenance. The application should be fast and responsive, with minimal lag times and quick loading speeds. This can be achieved through proper server and database optimization.

6. Other Requirements

- Multilingual support: Depending on the target audience of the website, it may be important to support multiple languages to ensure accessibility and reach a wider user base.
- User feedback: To continuously improve the website and better meet user needs, the website could collect feedback from users through ratings, and reviews.

Appendix A: Glossary

- Bugs: an unexpected defect, fault, flaw, or imperfection.
- Accessibility: the quality of being easy to obtain or use.
- Encryption: method by which information is converted into secret code that hides the information's true meaning.
- MERN Stack: stands for MongoDB, Express, React, Node, after the four key technologies that make up the stack. MongoDB — document database. Express(.js) — Node.js web framework. React(.js) — a client-side JavaScript framework.
- GitHub: an Internet hosting service for software development and version control using Git.